

第六讲：多视图几何/立体视觉



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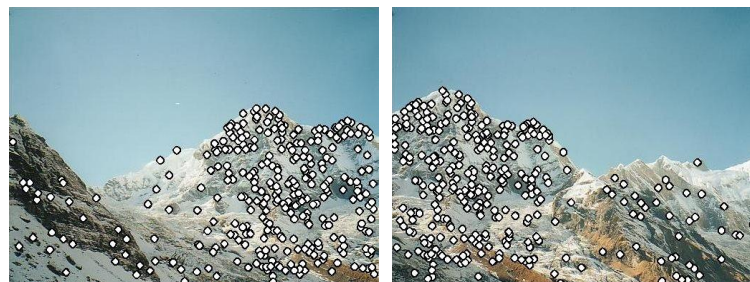
知识回顾：特征检测、描述与匹配



• 合成全景图的主要过程

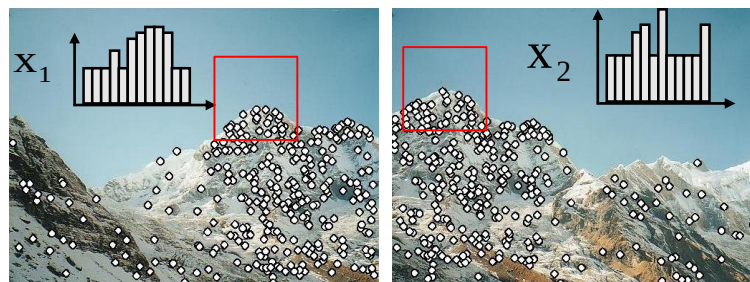
1, 特征检测

(Harris角点检测)



2, 特征描述

(SIFT)

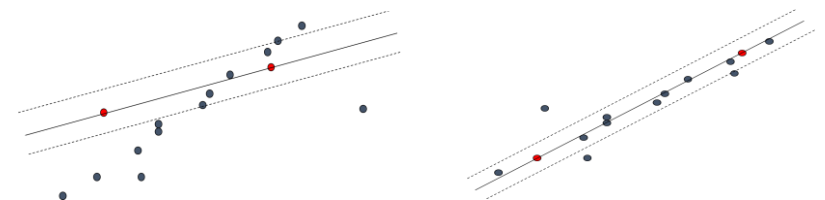


3, 特征匹配

(特征相似性计算)



4, (仿射) 变换参数拟合 (RANSAC)



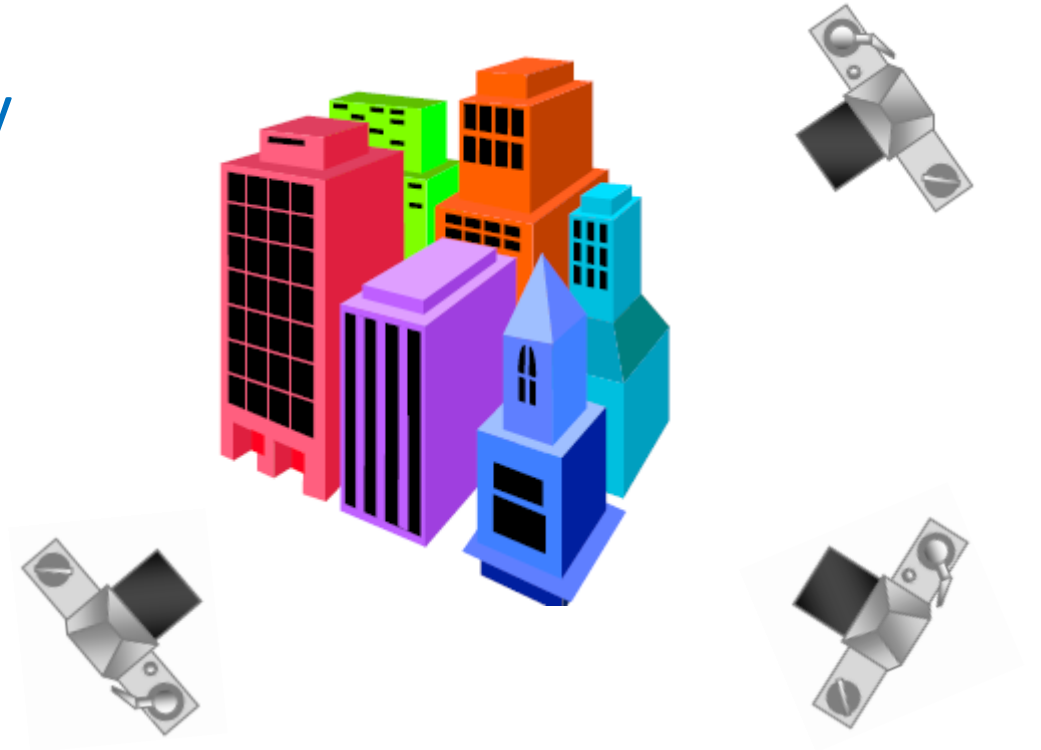
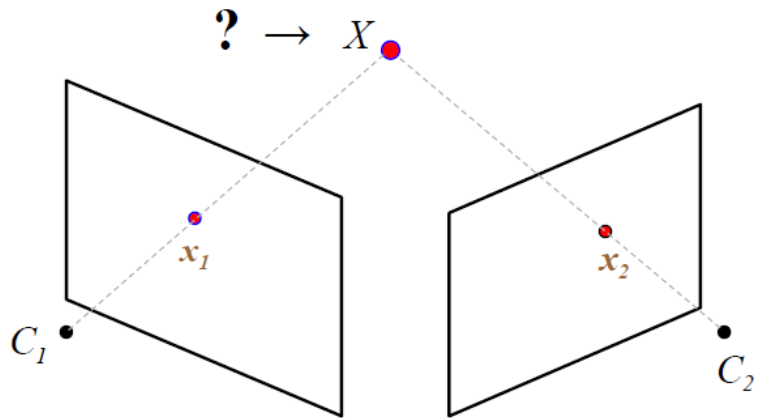
5, 图像对齐与融合



Multi-view Geometry 多视图几何



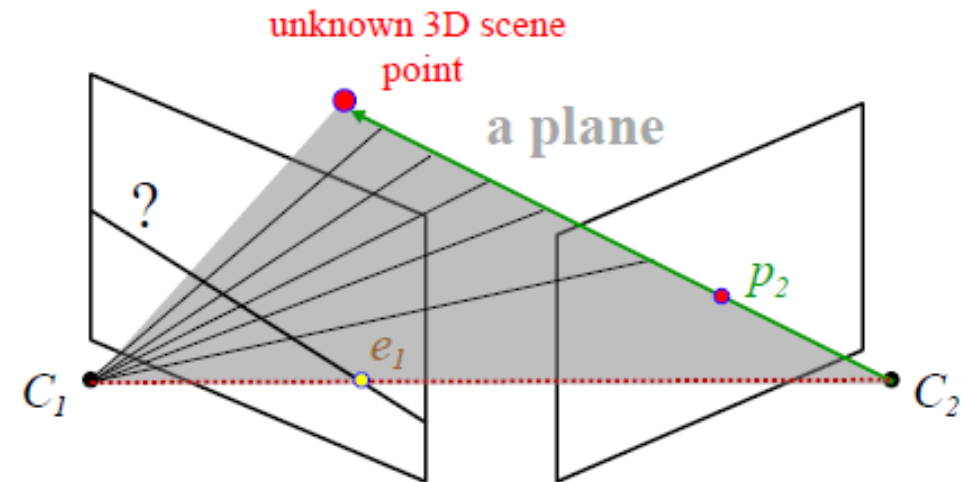
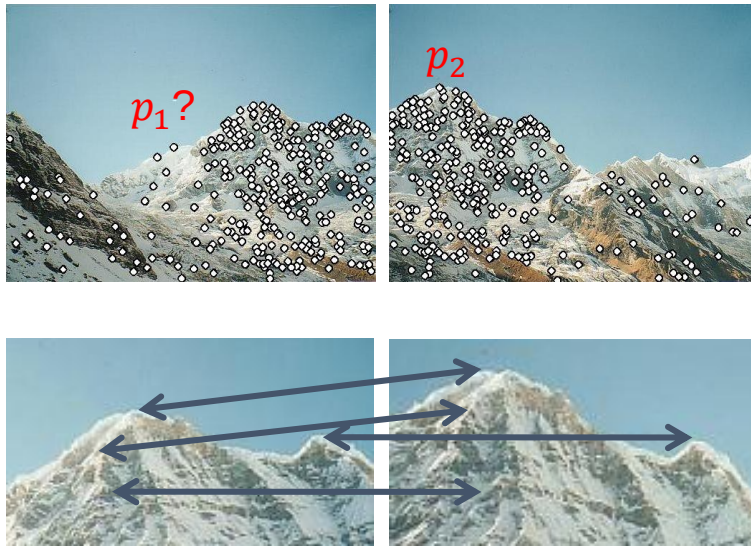
- Multi-view reconstruction: **shape from two or more images**
- To “triangulate” 三角测量
 - Should estimate camera viewpoints
- Need to learn about **multi-view geometry**



Multi-view Geometry 多视图几何

- Point matching

- For a point p_2 in the right image, do we need to search all pixels in the left image to find its corresponding point p_1 ?
- **Epipolar geometry (极线几何)** tells use p_1 is only on a certain line in the left image



Summary



- Projective Camera model 投影相机模型
 - Intrinsic and extrinsic parameters (相机的内参和外参)
 - Camera calibration (相机校正)
- Epipolar geometry (极线几何)
 - Essential matrix (本征矩阵)
 - Fundamental Matrix (基础矩阵)
 - Estimation of Essential and Fundamental matrices
- Application of multi-view geometry

相机内部参数

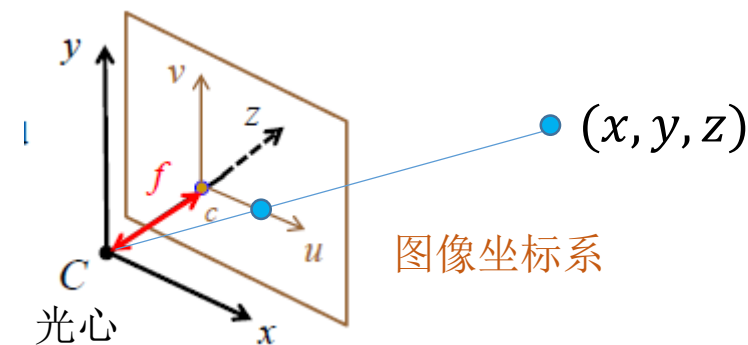
- Camera-centered 3D coordinate system
- A general model for camera
 - Different f_x and f_y
 - Skew coefficient s (斜切系数)

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = \begin{bmatrix} f_x & s & u_c \\ 0 & f_y & v_c \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Homogenous representation
齐次坐标

K
相机内参矩阵

相机坐标系中的点



In general, matrix K of intrinsic camera parameters is 3x3 upper triangular.
It has 5 degrees of freedom.

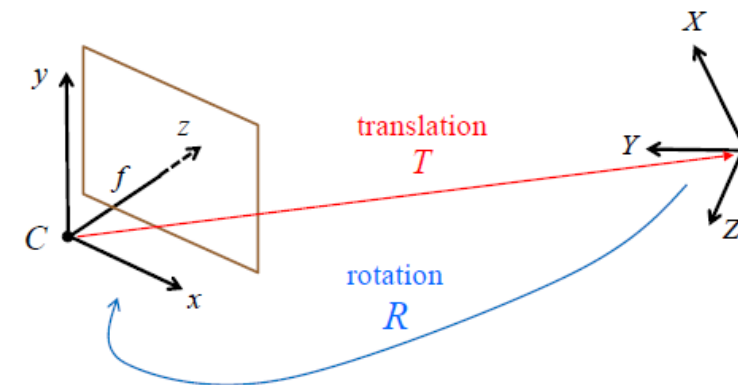
相机内部参数与外部参数

- 世界坐标系中的三维点映射到相机坐标系

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3 \times 1} = \underbrace{\begin{bmatrix} R & T \end{bmatrix}}_{3 \times 4} \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}_{4 \times 1}$$

- 相机坐标系中的三维点映射到图像中

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix}_{\text{homogeneous image coordinates}} = K \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{\text{camera-based 3D coordinates}}$$



- 世界坐标系中的三维点映射到图像中

Projection Matrix P
投影矩阵

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix}_{\text{homogeneous 2D image coordinates } \tilde{p}} = K \cdot \begin{bmatrix} R & T \end{bmatrix} \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}_{\text{homogeneous 3D world coordinates } \tilde{X}}$$

内部参数
外部参数

K: project (5 d.o.f.)

R: rotate (3 d.o.f.)

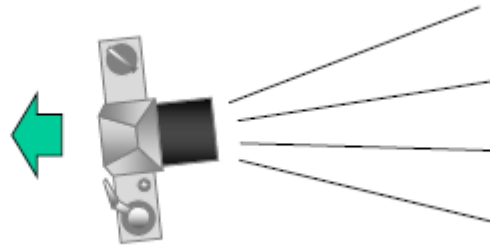
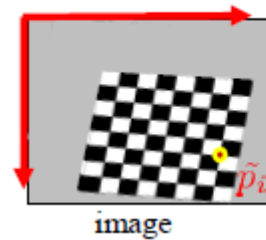
T: translate (3 d.o.f.)

如何估计相机的内部参数和外部参数值?

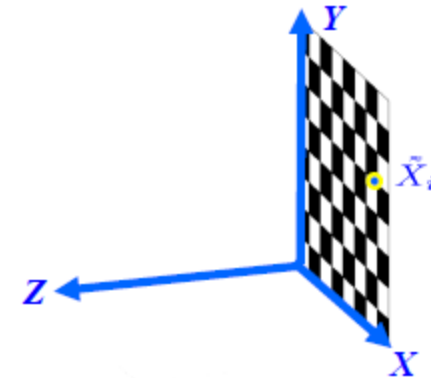
Camera calibration 相机校正

- Basic calibration technique
 - Assume a set of **3D points** $\{\tilde{X}_i\}$
 - with known world coordinates
 - and a set of matching **image points** $\{\tilde{p}_i\}$

图像坐标系



三维世界坐标系



用于相机校正的棋盘格

$$\{\tilde{X}_i\} \longleftrightarrow \{\tilde{p}_i\}$$

Find camera matrix **P** from known matches
Then, find intrinsic and extrinsic parameters
(use matrix factorization)

Camera calibration 相机校正

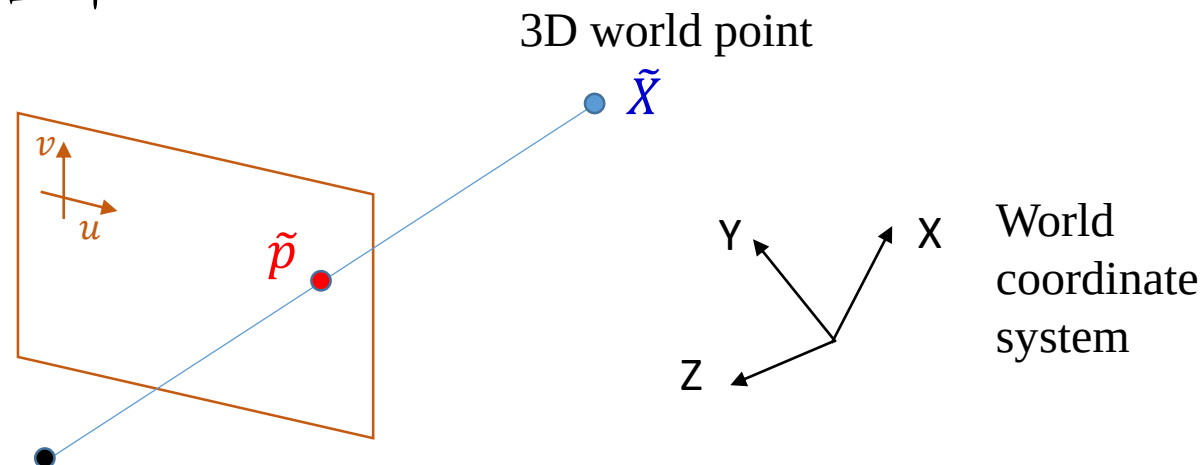


- 相机投影矩阵P的估计

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = K \cdot [R | T] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Projection matrix P

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & g & k & l \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$



- P has 12 entries, 11 dof
- Give pairs of $\{\tilde{X}_i\}$ and $\{\tilde{p}_i\}$

We want to solve for P

How many pairs are needed?

用6组以上的对应点，通过最小二乘法估计投影矩阵P

Use more than 6 matched pairs (least square problem)

Camera calibration 相机校正



- Intrinsic parameter estimation from P 基于P进一步估计相机内参K
 - Given P, how can we get K (as well as R, T)?

$$\overset{\text{已知}}{P} = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & g & k & l \end{bmatrix} = K \cdot \underbrace{[R \mid T]}_{\substack{\text{未知} \\ 3 \times 3 \quad 3 \times 4}}$$

$$K = \begin{bmatrix} f_x & s & u_c \\ 0 & f_y & v_c \\ 0 & 0 & 1 \end{bmatrix}$$

先验知识:

- K是上三角矩阵
- 旋转矩阵R是一个正交矩阵

Camera calibration 相机校正



- Intrinsic parameter estimation from P 基于P进一步估计相机内参K
- Matrix factorization 矩阵分解
 - Theorem: for any matrix A there is an orthogonal matrix Q and an upper (or right) triangular matrix R such that $A = \mathcal{R}Q$

$$P = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & g & k & l \end{bmatrix} = [\mathcal{R}Q \mid a] = \mathcal{R} \cdot [Q \mid \mathcal{R}^{-1}a]$$

$A = \mathcal{R}Q \mid a$

上三角矩阵 正交矩阵

$$P = K \cdot [R \mid T]$$

内参矩阵K 旋转矩阵R
= 上三角矩阵R = 正交矩阵Q

将 P 进行上三角分解，并归一化，即可得到 K

Camera calibration 相机校正



- Camera normalization (calibrated camera) 归一化相机
- When intrinsic parameters K are known
 - Switch to a new image coordinate system $(\tilde{u}, \tilde{v})$

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = K \cdot [R | T] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} w\tilde{u} \\ w\tilde{v} \\ w \end{bmatrix} = K^{-1} \begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = [R | T] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

This corresponds to a transform of the camera's image

Then, the camera's new projection matrix $\tilde{P}$ becomes

$$\tilde{P} = [R | T]$$

内参为单位矩阵，投影矩阵仅包含外部参数

Camera calibration 相机校正



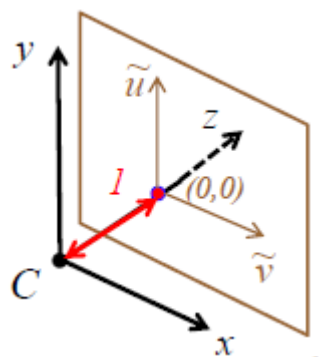
- Camera normalization (calibrated camera) 归一化相机
 - After normalization, the intrinsic parameters corresponds to an identity matrix

$$[R | T] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot [R | T]$$
$$\tilde{K} = I$$



Focal length $f = 1$

Image offset $(u, v) = (0, 0)$



归一化相机的相机坐标系

- 以相机位置为坐标原点，焦距为1
- 图像中心坐标定义为光心

归一化后，图像与相机内部参数无关，仅取决于外部参数

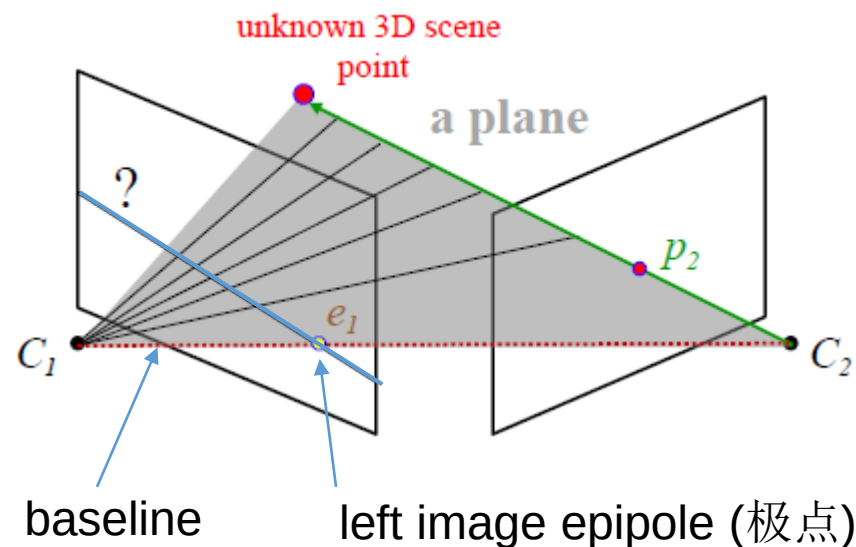
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Epipolar geometry 极线几何

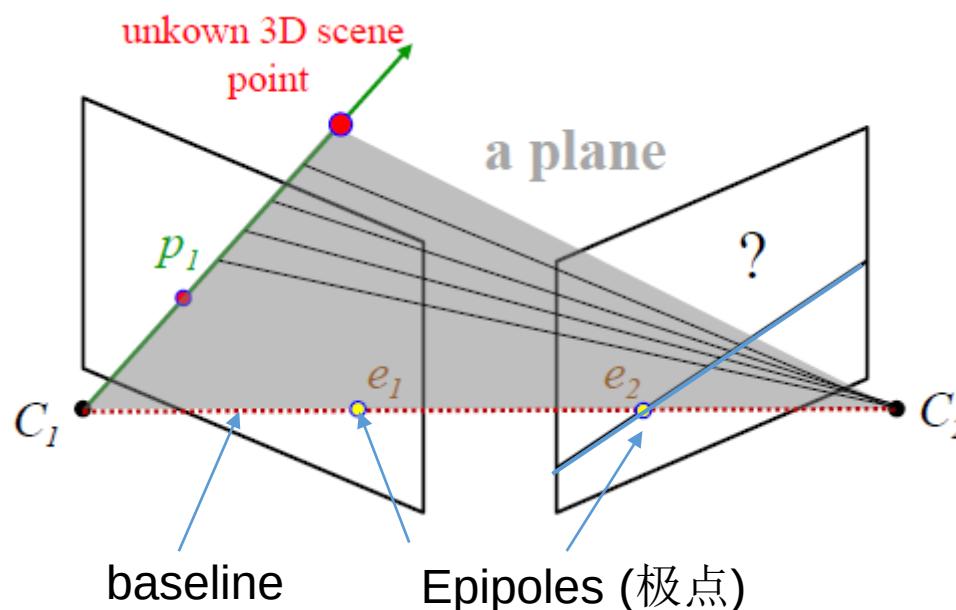
- Epipolar line (极线)
 - p_2 's corresponding point must be on this line in the left image



Any right image point p_2 corresponds to some left image epipolar line.
It is a projection of ray $C_2 \rightarrow p_2$

Epipolar geometry 极线几何

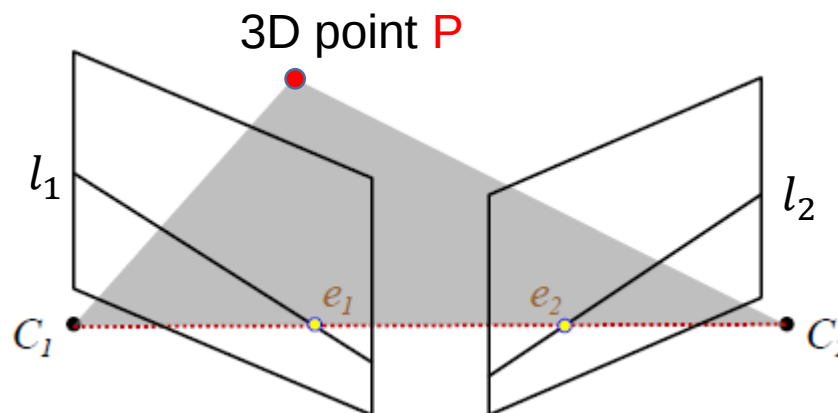
- Epipolar line example 极线的例子
 - **Similarly**, for any given point p_1 in the left image
 - p_1 's corresponding point must be on this line in the right image



Note p_1 , C_1 , C_2 , and the epipolar lines are in the same plane.

Epipolar geometry 极线几何

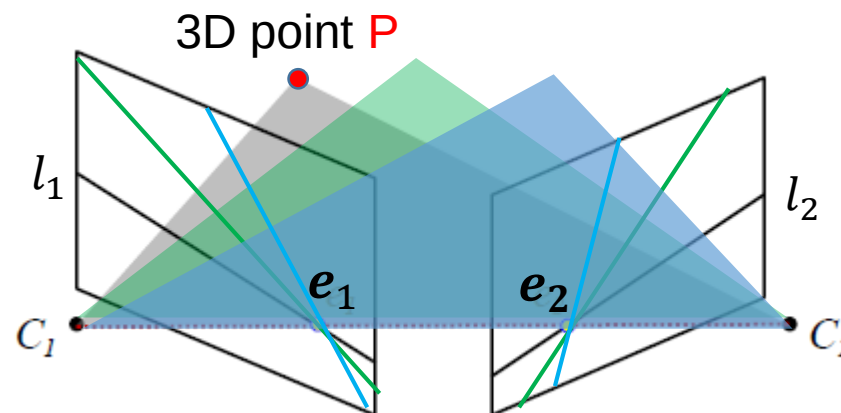
- Corresponding epipolar lines (极线的对应关系)
 - For a given 3D point, there is a pair of epipolars l_1 and l_2
 - P , l_1 and l_2 , C_1 and C_2 are in the same plane (极平面)



Epipolar geometry 极线几何



- Corresponding epipolar lines (极线的对应关系)
 - For a given 3D point, there is a pair of epipolars l_1 and l_2
 - P , l_1 and l_2 , C_1 and C_2 are in the same plane (极平面)



C_1 和 C_2 是一系列极平面的相交线

Intersection of epipolar planes with image planes define a system of epipolar lines

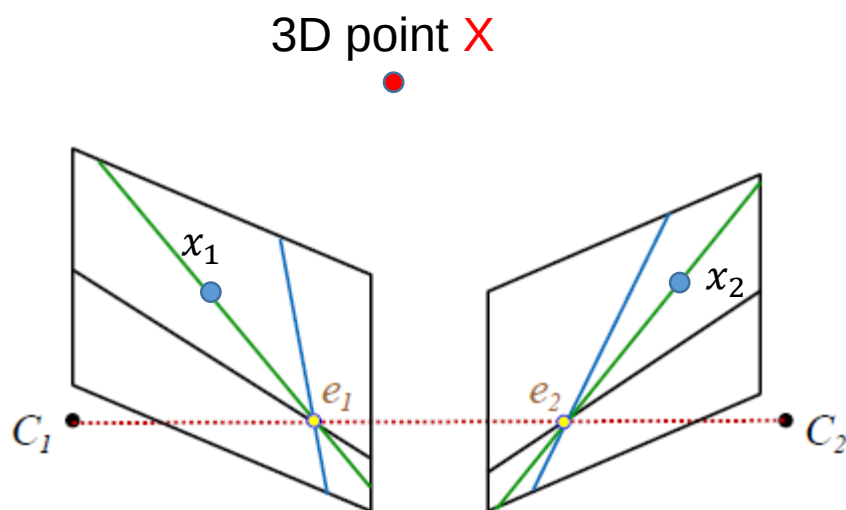
Corresponding points can be only on corresponding epipolar lines

- important to know such lines when searching for corresponding pairs of points

Epipolar geometry 极线几何



- Corresponding epipolar lines (极线的对应关系)



如何在 P_1 和 P_2 未知的情况下，找到 x_1 和 x_2 的变换关系？

How can we compute epipolar lines for a given pair of images?

- If known, camera projection matrices P_1 and P_2 contain all information

$$e_1 = P_1 C_2 \quad e_2 = P_2 C_1 \quad x_1 = P_1 X \quad x_2 = P_2 X$$

- What if only relative position of two cameras?
 - Can estimate a single 3x3 **essential matrix** rather than two 3x4 matrices $P = (R|T)$

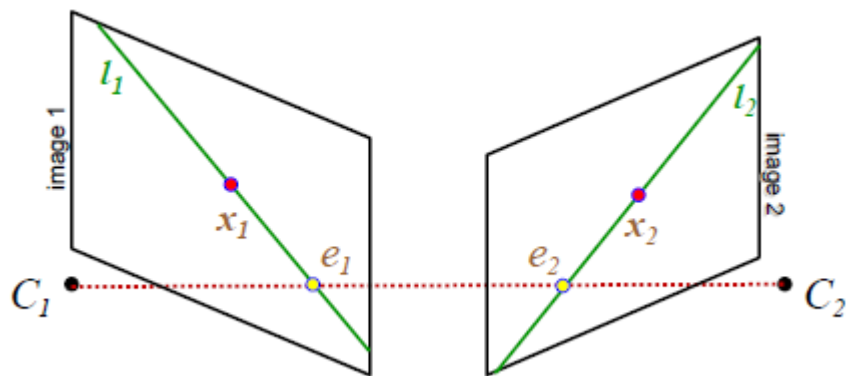
Epipolar geometry 极线几何



- Essential Matrix E (本征矩阵)

- The corresponding epipolar lines is fully described by a 3x3 matrix E

对于归一化相机，两幅图像中的对应点可以通过以下关系式得到：



$$x_2^T E x_1 = 0$$

for any pair of pixels/points x_1 and x_2
on the corresponding epipolar lines
(assuming calibrated cameras)

Given x_1 in image 1, vector $l_2 = E x_1$ gives equation $x_2 \cdot l_2 = 0$

Given x_2 in image 2, vector $l_1 = E x_2$ gives equation $x_1 \cdot l_1 = 0$

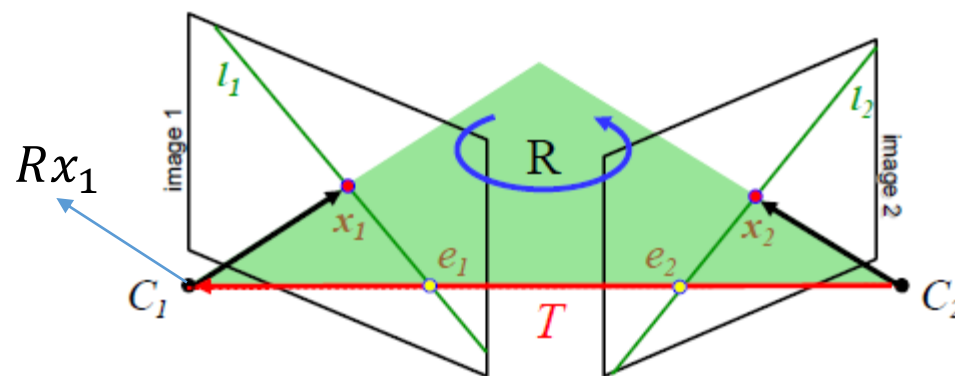
Epipolar geometry 极线几何

- Essential Matrix E (本征矩阵)

- Assuming calibrated cameras
- x_1 and x_2 can be treated as 3D points (vectors) in the camera-centered coordinates

$$P_1 = [I | 0]$$

$$P_2 = [R | T]$$



R and T convert camera 1 (world) coordinates to camera 2

vectors x_1 , x_2 and T are on the same plane

$$x_2 \cdot [T \times (Rx_1)] = 0$$

dot product cross product

NOTE: Rx_1 is vector x_1 in camera 2 coordinates and $T \cdot Rx_1$ is the green plane's normal (camera 2 coordinates)

Epipolar geometry 极线几何



- Essential Matrix E (本征矩阵)

Note: cross product $a \times b$ can be represented as matrix multiplication

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, \quad a \times b = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$
$$a \times b = \underbrace{\begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}}_{\text{notation } [a]_{\times}} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$
$$a \times b = [a]_{\times} b$$

$$x_2 \cdot [T \times (R x_1)] = 0$$

dot product cross product



Matrix expression

$$x_2^T [T]_{\times} R x_1 = 0$$
$$x_2^T E x_1 = 0$$

$$E = [T]_{\times} R$$

Epipolar geometry 极线几何



- Fundamental Matrix F (基础矩阵)

Remember: we have assumed calibrated camera coordinates for E $x_2^T E x_1 = 0$

$$\tilde{x} = K^{-1}x$$

归一化相机中的
图像坐标系

原始图像
坐标系

$$\underbrace{x_2^T}_{\tilde{x}_2^T} \underbrace{K^{-T} E K^{-1}}_{\tilde{x}_1} x_1 = 0$$

对于未归一化的相机:

$$x_2^T F x_1 = 0 \quad F = K^{-T} E K^{-1}$$

F : fundamental matrix

Define epipolar lines for uncalibrated cameras

Epipolar geometry 极线几何



- 基础矩阵 F 的性质

- 如果 F 为基础矩阵, 那么 kF 也为基础矩阵
- F 的秩为2 $\text{rank}(F) = 2$

} 减去两个自由度, F 有7个自由度

$$x_2^T F x_1 = 0 \quad F = K^{-T} E K^{-1}$$

$$E = [T]_{\times} R$$

秩为2

$$[T]_{\times} = \begin{pmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & T_x & 0 \end{pmatrix} = \begin{bmatrix} A_1 & A_2 & A_3 \end{bmatrix}$$

第三列 A_3 可以用第一列 A_1 和第二列 A_2 线性表示

$$A_3 = -\frac{T_x}{T_z} A_1 - \frac{T_y}{T_z} A_2$$

Epipolar geometry 极线几何



- Essential and Fundamental Matrices

Essential matrix E 本征矩阵

- Epipolar lines $x_2^T E x_1 = 0$
(for calibrated cameras)

$$E = [T]_{\times} R$$

- 5 dof (6 from R and T - scale of T)

Fundamental matrix F 基础矩阵

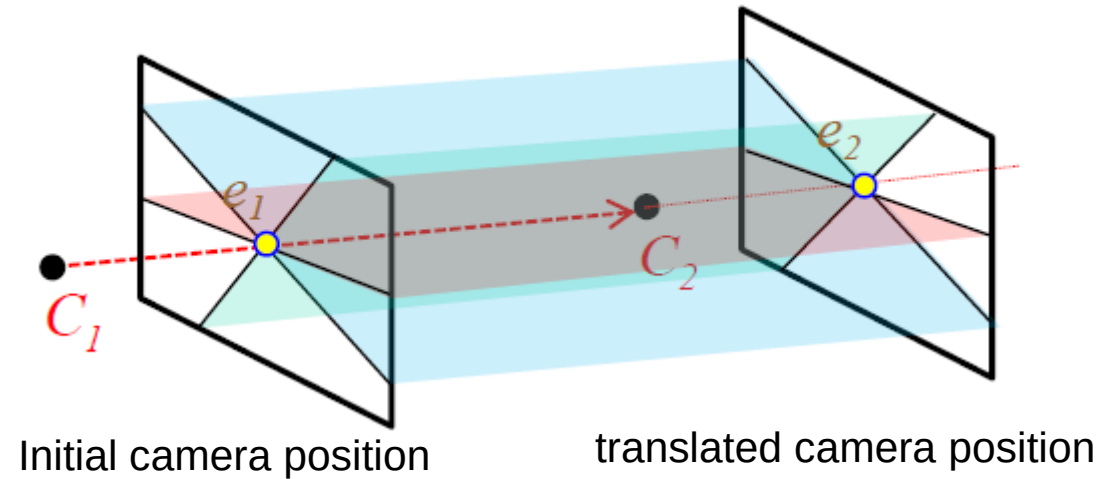
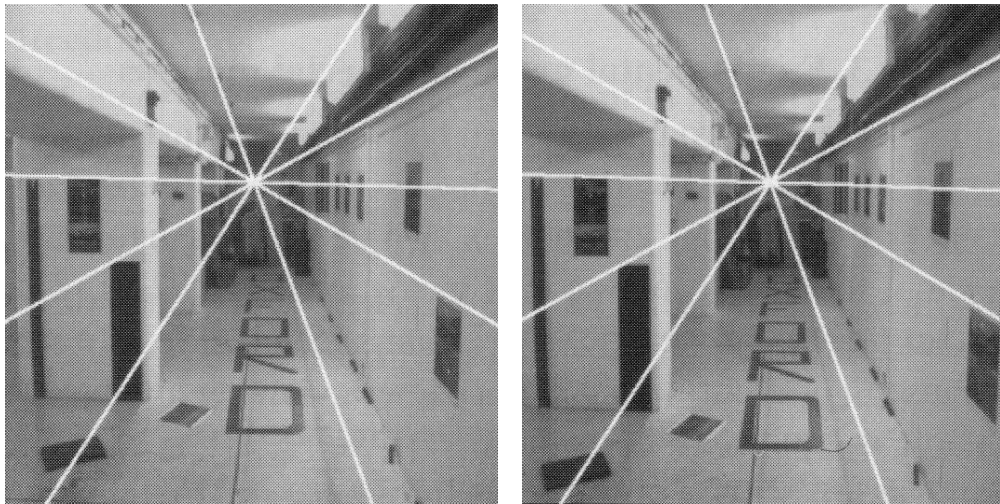
- Epipolar lines $x_2^T F x_1 = 0$
(for uncalibrated cameras)

$$F = K^{-T} E K^{-1}$$

- 7 dof (9 par., -scale & $\det F=0$)

Epipolar geometry

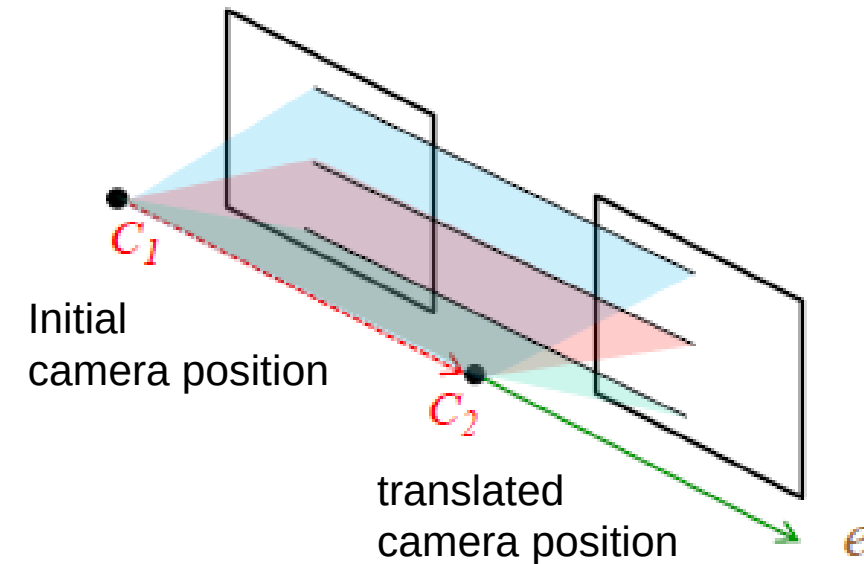
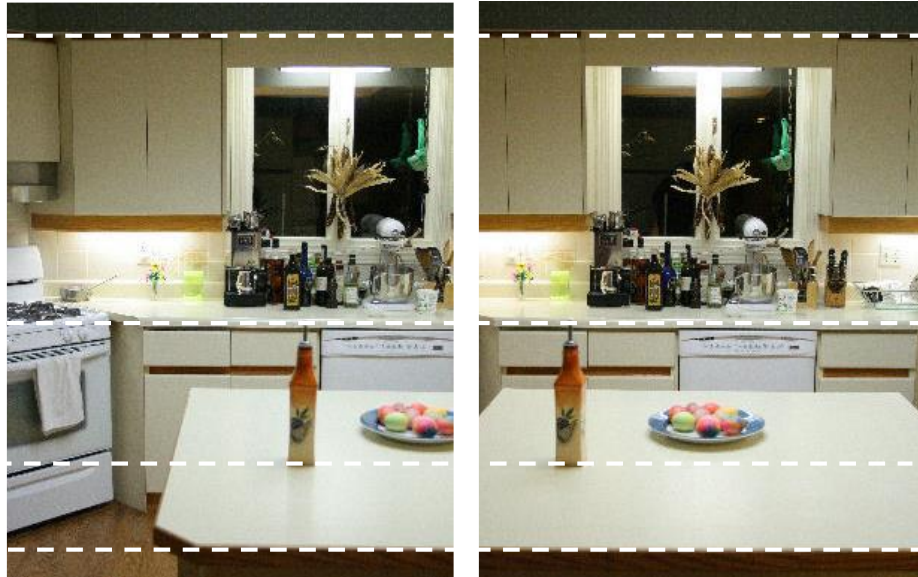
- Example: Camera translation
 - Forward translation



相机向前移动，两幅图像的极点位置相同，且配对的极线在图像中也具有同样的位置

Epipolar geometry

- Example: Camera translation
 - Panning camera motion



- Epipole $e_1 = e_2 = e$ is a point at infinity for the image plane
- The corresponding epipolar lines are parallel lines

Estimation of E and F from $N \geq 8$ matches



- 8-point method 求基础矩阵 F 的“8点算法”
 - Assume corresponding points $\mathbf{x}_i$ and $\bar{\mathbf{x}}_i$ in two images
 - They must lie on the corresponding epipolar lines, thus

$$\bar{\mathbf{x}}_i^T F \mathbf{x}_i = 0 \quad (\text{Use } E \text{ for calibrated images})$$

Denote $\mathbf{x}_i = (x_i, y_i, z_i)$ and $\bar{\mathbf{x}}_i = (\bar{x}_i, \bar{y}_i, \bar{z}_i)$, then

$$\begin{aligned} \bar{\mathbf{x}}_i^T F \mathbf{x}_i = & F_{11}\bar{x}_i x_i + F_{12}\bar{x}_i y_i + F_{13}\bar{x}_i z_i \\ & + F_{21}\bar{y}_i x_i + F_{22}\bar{y}_i y_i + F_{23}\bar{y}_i z_i \\ & + F_{31}\bar{z}_i x_i + F_{32}\bar{z}_i y_i + F_{33}\bar{z}_i z_i = 0 \end{aligned}$$

One pair $\mathbf{x}_i, \bar{\mathbf{x}}_i$ gives only one linear equation

Eight is enough to determine elements of 3x3 matrix F (scale is arbitrary)

Estimation of E and F from $N \geq 8$ matches



- In matrix form

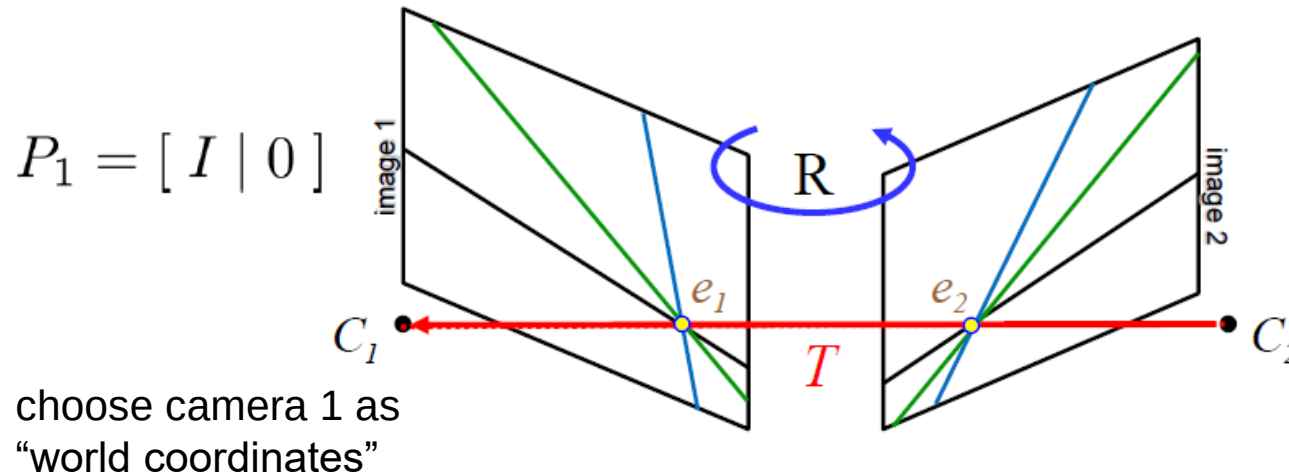
- One row for each of $N \geq 8$ matches

$$\begin{matrix} & N \times 9 & & 9 \times 1 & & N \times 1 \\ & \begin{matrix} \bar{x}_1 x_1 & \bar{x}_1 y_1 & \bar{x}_1 z_1 & \cdots & \bar{z}_1 z_1 \end{matrix} & & \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ \vdots \\ F_{33} \end{bmatrix} & = & \begin{bmatrix} 0 \\ \vdots \end{bmatrix} \\ \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} & & & & & \\ & \mathbf{A} & & \mathbf{f} & & \end{matrix}$$

$$\mathbf{A}\mathbf{f} = 0$$

Extracting cameras from essential matrix E

- Now assume essential matrix E is given, need to find P_1 and P_2



Theorem: 3x3 matrix E is *essential* ($\exists R, T : E = [T]_{\times} R$) if and only if two of its singular values are equal, and the third is zero.
[H&Z. Sec 9.6 p.257]

SVD for essential matrix E is:

$$E = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} V^T$$

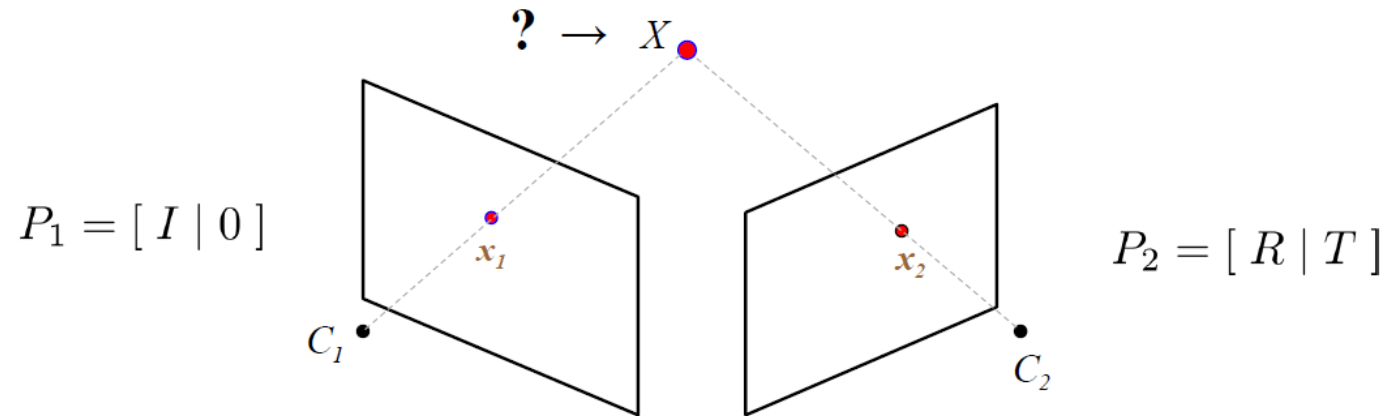
Given E , find R and T such that

$$E = [T]_{\times} R$$

Triangulation 三角测量



- Obtain 3D point X , given
 - Projection matrices P_1 and P_2 , and a match $(\mathbf{x}_1, \mathbf{x}_2)$



Projection constraints

$$\begin{bmatrix} w_1 u_1 \\ w_1 v_1 \\ w_1 \end{bmatrix} = P_1 \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad \begin{bmatrix} w_2 u_2 \\ w_2 v_2 \\ w_2 \end{bmatrix} = P_2 \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

6 equations with 5 unknown (X, Y, Z, w_1, w_2)

But, we do not care about w_1 and w_2

$\Rightarrow$ 4 equations with 3 unknown (X, Y, Z)

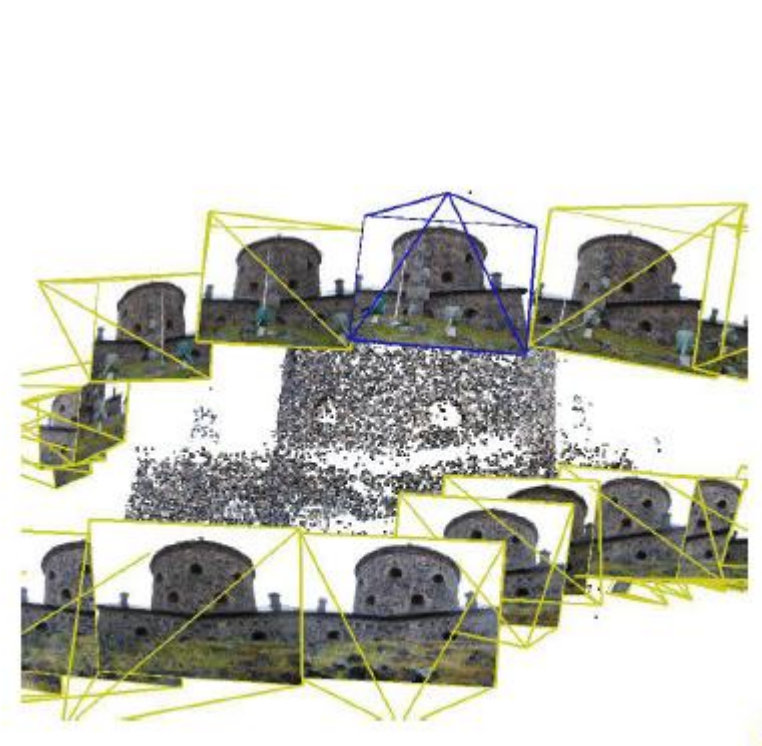
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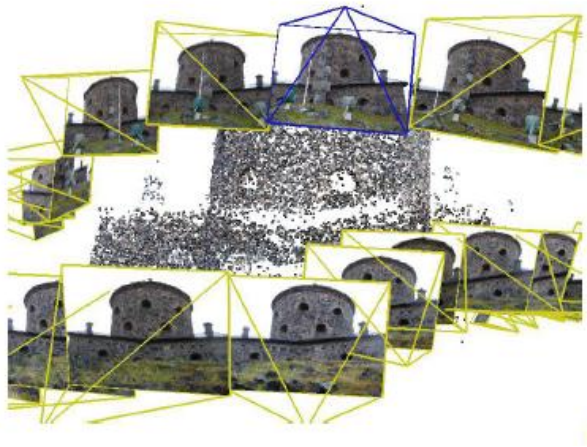
Structure from motion

- Obtain 3D reconstruction when moving the camera



Structure from motion workflow

- Basic sequential reconstruction
 - For the **first two images**, use 8-point algorithm to estimate essential matrix E , cameras, and triangulate some points $\{X_i\}$.
 - Each **new view** should see some previously reconstructed scene points $\{X_i\}$, Use such points to estimate new camera position.
 - Add new scene points using triangulation.
 - If there are more cameras, iterate previous two steps.



Dense 3D reconstruction

- Computing surfaces in 3D Volumes (volumetric reconstruction)



multiple wide baseline views

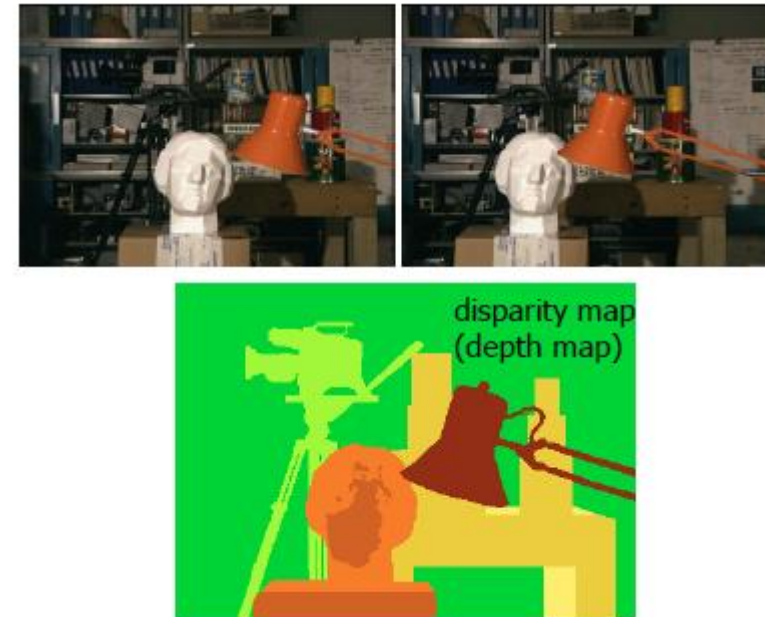


Dense 3D reconstruction

- Computing dense Depth Maps (stereo)



Two narrow wide baseline views



Application of multi-view geometry



- Rigid motion estimation
 - We were fitting a single essential/fundamental matrix to a pair of images corresponding to two different view points
 - **Q**: Can matched features in two images support more than one fundamental matrix?

能否从两张图像中估计多个基础矩阵？

Hint: we assumed that the scene is stationary and only camera moved or, equivalently, that the camera is stationary but the whole 3D scene moved (R,T).

Application of multi-view geometry

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**multi-model fitting with
fundamental matrices**

Application of multi-view geometry

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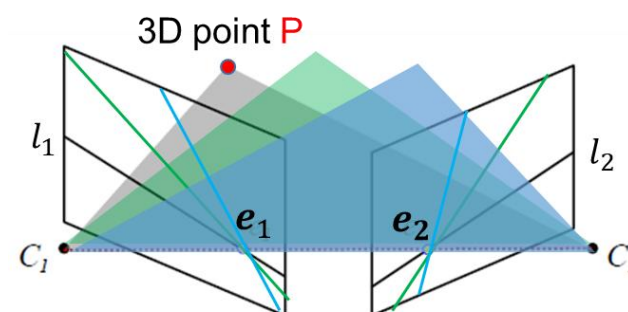
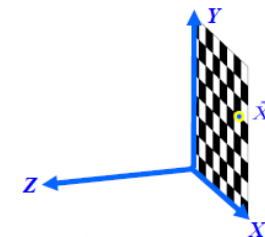
**multi-model fitting with
fundamental matrices**

Application of multi-view geometry

- Augmented Reality 增强现实
 - if camera position C and orientation R are known (in addition to K) then can insert “new” objects into the 3D scene
 - particularly useful for movies: camera path can be computed
 - can generate correct views of new objects



- 相机的校正
 - 通过棋盘格校正得到投影矩阵P
 - 对P进行对角化得到内参矩阵K
- 极线几何
 - 两幅图像中对应的点通过极线关联
 - 本征矩阵E
 - 基础矩阵F
 - 重建三维坐标：三角测量
- 应用
 - 三维重建、运动估计、增强现实等



$$x_2^T E x_1 = 0$$

$$x_2^T F x_1 = 0$$

$$F = K^{-T} E K^{-1}$$

